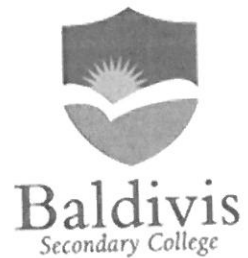


Year 10  
Pathway 2  
Test 2  
2018



Name: SOLUTIONS.

Class: \_\_\_\_\_

Time: 35 minutes

Total Mark: \_\_\_\_/35

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 A4 page of notes (1 side), Calculator allowed.

**Question 1**

**10 marks**

Solve each of the following equations (that is, find the value for the pronumeral that makes each statement true).

a)  $x + 2 = 8$   
 $\begin{array}{r} -2 \quad -2 \\ x + 2 = 8 \\ \hline x = 6 \end{array}$  ①

b)  $\frac{6k}{6} = \frac{36}{6}$   
 $k = 6$  ①

c)  $\frac{w}{5} \times 5 = -9 \times 5$   
 $w = -45$  ①

d)  $5a + 6 = 26$   
 $\begin{array}{r} -6 \quad -6 \\ 5a + 6 = 26 \\ \hline 5a = 20 \\ \hline a = 4 \end{array}$  ①

e)  $\frac{f}{4} + 6 = 16$   
 $\begin{array}{r} -6 \quad -6 \\ \frac{f}{4} + 6 = 16 \\ \hline \frac{f}{4} = 10 \\ \hline f = 40 \end{array}$  ①

f)  $\frac{x}{2} - \frac{3x}{5} = \frac{1}{4}$   
 $\begin{array}{r} \frac{10x}{20} - \frac{12x}{20} = \frac{5}{20} \\ \hline \frac{10x - 12x}{20} = \frac{5}{20} \\ \hline \frac{-2x}{20} = \frac{5}{20} \\ \hline -2x = 5 \\ \hline x = -2\frac{1}{2} \end{array}$  ①

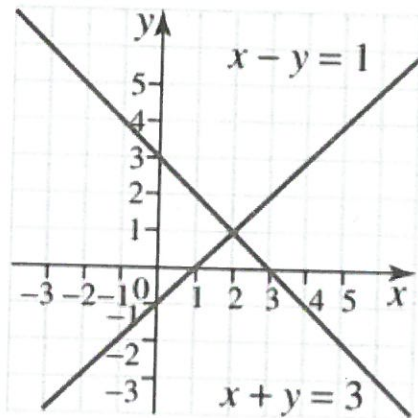
### Question 2

2 marks

Use the graphs below of the given simultaneous equations to write the point of intersection and, hence, the solution of the simultaneous equation.

Equation 1:  $x - y = 1$

Equation 2:  $x + y = 3$



$x - y = 1$  Equation 1  
 $x + y = 3$  Equation 2

Equation 1:  $x - y = 1$   
 $2 - 1 = 1$   
LHS = RHS (✓)

Equation 2:  $x + y = 3$   
 $2 + 1 = 3$   
LHS = RHS (✓)

Point of intersection:  
(2, 1) (✓)

In both equations LHS = RHS therefore the solution (2, 1) is correct

### Question 3

5 marks

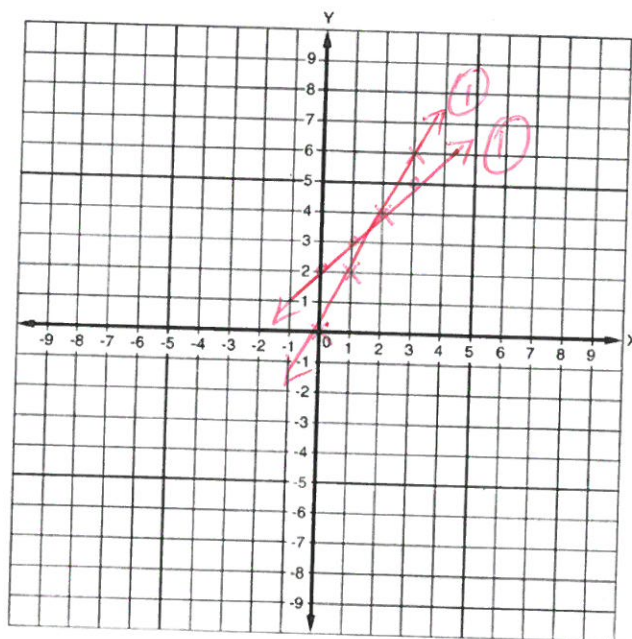
Plot the following linear graphs for the given range of  $x$  - values on the same Cartesian plane:

a)  $y = x + 2$

b)  $y = 2x$

|   |   |   |   |   |
|---|---|---|---|---|
| x | 0 | 1 | 2 | 3 |
| y | 2 | 3 | 4 | 5 |

|   |   |   |   |   |
|---|---|---|---|---|
| x | 0 | 1 | 2 | 3 |
| y | 0 | 2 | 4 | 6 |



c) Use these graphs to determine the point of intersection and, hence, the solution of the simultaneous equations.

(2, 4) (✓)

Question 4

5 marks

Solve the following pair of simultaneous equations using a graphical method

a.  $y = 2x + 3$

b.  $y = \frac{1}{2}x + 1$

$$y = 2x + 3$$

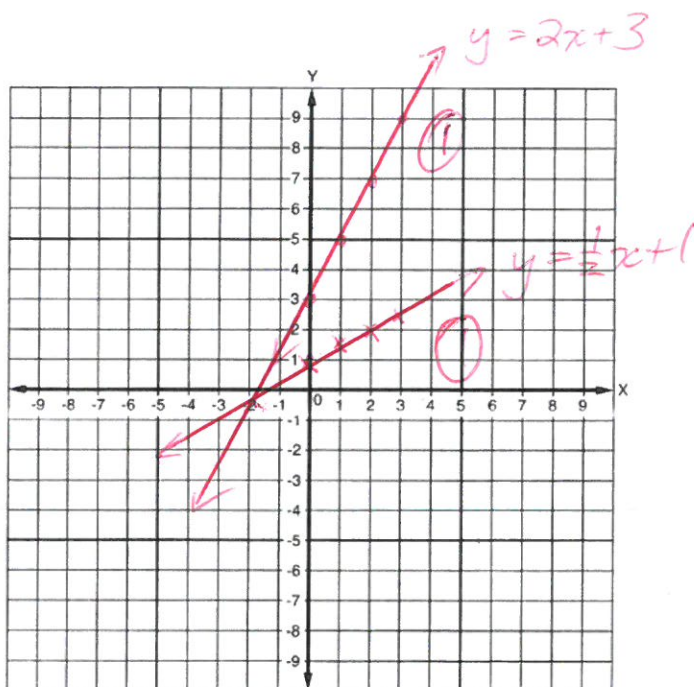
|   |   |   |   |   |
|---|---|---|---|---|
| x | 0 | 1 | 2 | 3 |
| y | 3 | 5 | 7 | 9 |

$$y = \frac{1}{2}x + 1$$

|   |   |     |   |     |
|---|---|-----|---|-----|
| x | 0 | 1   | 2 | 3   |
| y | 1 | 1.5 | 2 | 2.5 |

$y = 2x + 3$   
 $c = 3$

$y = \frac{1}{2}x + 1$   
 $c = 1$



- c. Use these graphs to determine the point of intersection and, hence, the solution of the simultaneous equations.

$(-2, -0.5)$

①

Question 5

6 marks

Solve the following pair of simultaneous equations:

$$y = 5x - 8$$

$$y = -3x + 16$$

Find  $x$

$$y = 5x - 8 \text{ - Equation 1}$$

$$y = -3x + 16 \text{ - Equation 2}$$

$$5x - 8 = -3x + 16 \quad (1)$$

$$\begin{array}{r} +3x \\ 8x - 8 = 16 \\ +8 \quad +8 \end{array} \quad (2)$$

$$\frac{8x}{8} = \frac{24}{8} \quad (3)$$

$$x = 3 \quad (4)$$

Find  $y$

$$y = 5x - 8 \text{ - Equation 1}$$

$$y = -3x + 16 \text{ - Equation 2}$$

$$\begin{array}{l} y = 5x - 8 \\ = 5 \times 3 - 8 \end{array}$$

$$y = 7 \quad (5)$$

$$y = -3x + 16$$

$$y = -3 \times 3 + 16$$

$$y = -9 + 16$$

$$y = 7 \quad (6)$$

$$\text{Solution: } x = 3 \quad y = 7 \quad (3, 7) \quad (7)$$

Question 6

3 marks

Decide if the graph of each pair of rules will be parallel, perpendicular or neither

a.  $y = -3x - 8$  and  $y = \frac{1}{3}x + 1$

$$y = -3x - 8 \quad m = -3$$

$$y = \frac{1}{3}x + 1 \quad m = \frac{1}{3}$$

$$\text{Test } m_1 \times m_2 = -1 \quad (8)$$

$$-3 \times \frac{1}{3} = -1 \quad (9)$$

Both lines are perpendicular  $(10)$

b.  $y = \frac{1}{2}x + 2$  and  $y = \frac{1}{2}x - 4$

$$y = \frac{1}{2}x + 2 \quad m = \frac{1}{2} \quad (11)$$

$$y = \frac{1}{2}x - 4 \quad m = \frac{1}{2} \quad (12)$$

$$\text{Test } m_1 \times m_2 = -1$$

$$\frac{1}{2} \times \frac{1}{2} = 0.25$$

Not perpendicular

They have the same gradient (0.5) so they are parallel  $(13)$

## Question 7

4 marks

Find the equation of the straight line that passes through the point (4, -1) and is parallel to the straight line with equation  $y = 2x - 5$

$$y = mx + c$$

$$y = 2x - 5 \quad m = 2$$

①

$$\text{so } y = 2x + c$$

Find  $c$  by substituting (4, -1)

$$-1 = 2 \times 4 + c$$

①

$$-1 = 8 + c$$

$$-1 - 8 = c$$

$$-9 = c$$

①

$$\therefore y = 2x - 9$$

①

this equation has a  $m = 2$

so the equation that is parallel is  $y = 2x - 9$ .

END OF TEST

